Playfair Cipher

requirements:

- The input of plaintext that consists all 4 rules
- Generate the key square 5 x 5 which is not redundant with other team

Solution:

plaintext: "BATTLEHPX"

keyword: QUICKER

Step 1: create the key square

the key square 5 x 5 at the right side

Q	U	I/J	C	K
E	R	A	B	D
F	G	H	L	M
N	O	P	S	T
V	W	X	Y	Z

Step 2: Divide the plaintext into pairs

we have plaintext = BATTLEHPX

BA, TT, LE, HP, X2

Step 3. Apply Playfair rules

1 – Same row and same column : change the second index of pair into letter 'x', e.g. TT → TX

2 - Different row and different column : get an encrypted letter by the row intercept with column on the key square table.

3 – Same row and different column : get an encrypted letter by take the letter in the next index of that pair on the key square table.

4 – Diffrent row and same column : —11—

Pair	Rule	Encrypted Pair
BA	3	DB
TT	1	PZ
LE	2	FB
HP	4	PX
X2	3	YV

Q	U	I/J	C	K
E	R	A	B	D
F	G	H	L	M
N	O	P	S	T
V	W	X	Y	Z

Therefore, ciphertext = DBPZFBPXYV

+ Decrypted

ciphertext = DBPZFBPXYV

divided in to pair : DB, PZ, FB, PX, YV

Pair	role	Decrypted Pair
DB	3	BA
PZ	1	TX → TT
FB	2	LE
PX	4	HP
YV	3	X2

so, plaintext pair : BA, TT, LE, HP, X2

⇒ plaintext = BATTLEHPX